```
create database assignment ;
use assignment ;
CREATE TABLE Employees (
EmpID INT PRIMARY KEY,
Name VARCHAR(50) NOT NULL,
Dept VARCHAR(20) NOT NULL,
Role VARCHAR(20) NOT NULL,
Salary INT NOT NULL,
Gender CHAR(1) NOT NULL,
JoinDate DATE NOT NULL,
Address VARCHAR(100) NOT NULL,
City VARCHAR(50) NOT NULL,
State CHAR(2) NOT NULL,
Region VARCHAR(10) NOT NULL
);

INSERT INTO Employees (EmpID, Name, Dept, Role, Salary, Gender, JoinDate, Address, City, State,
Region) VALUES
(1, 'Alice', 'HR', 'Manager', 55000, 'F', '2019-03-15', '101 Maple St', 'New York', 'NY',
'East'),
(2, 'Bob', 'IT', 'Developer', 72000, 'M', '2020-07-22', '22 Oak Ave', 'Chicago', 'IL',
'Midwest'),
(3, 'Carol', 'Finance', 'Analyst', 68000, 'F', '2018-11-01', '303 Pine Rd', 'New York', 'NY',
'East'),
(4, 'David', 'HR', 'Executive', 52000, 'M', '2021-01-10', '44 Birch Ln', 'Chicago', 'IL',
'Midwest'),
(5, 'Eve', 'IT', 'Senior Dev', 85000, 'F', '2017-06-30', '55 Cedar Blvd', 'Houston', 'TX',
'South'),
(6, 'Frank', 'Finance', 'Manager', 74000, 'M', '2019-09-14', '606 Elm St', 'New York', 'NY',
'East'),
(7, 'Grace', 'Marketing', 'Executive', 60000, 'F', '2022-02-28', '77 Spruce Dr', 'Chicago', 'IL',
```

```
'Midwest'),
(8, 'Hank', 'IT', 'Lead', 90000, 'M', '2016-04-05', '88 Walnut Way', 'Houston', 'TX',
'South'),
(9, 'Ivy', 'HR', 'Executive', 58000, 'F', '2020-12-19', '99 Ash Ct', 'New York', 'NY',
'East'),
(10, 'Jack', 'Finance', 'Analyst', 65000, 'M', '2021-03-07', '10 Willow Pl', 'Chicago', 'IL',
'Midwest'),
(11, 'Karen', 'Marketing', 'Manager', 62000, 'F', '2018-08-23', '11 Poplar St', 'Houston', 'TX',
'South'),
(12, 'Leo', 'IT', 'Developer', 78000, 'M', '2019-11-11', '12 Sycamore Ave', 'New York', 'NY',
'East'),
(13, 'Mia', 'HR', 'Executive', 53000, 'F', '2022-05-16', '13 Hickory Rd', 'Chicago', 'IL',
'Midwest'),
(14, 'Nate', 'Finance', 'Manager', 71000, 'M', '2017-12-02', '14 Chestnut Ln', 'Houston',
'TX', 'South'),
(15, 'Olivia', 'Marketing', 'Analyst', 66000, 'F', '2020-06-18', '15 Magnolia Dr', 'New York',
'NY', 'East'),
(16, 'Paul', 'IT', 'Senior Dev', 82000, 'M', '2018-03-27', '16 Dogwood Blvd', 'Chicago', 'IL',
'Midwest'),
(17, 'Quinn', 'HR', 'Executive', 57000, 'F', '2021-09-09', '17 Hawthorn St', 'Houston', 'TX',
'South'),
(18, 'Rita', 'Finance', 'Lead', 76000, 'F', '2019-01-25', '18 Redwood Ave', 'New York', 'NY',
'East'),
(19, 'Sam', 'Marketing', 'Executive', 63000, 'M', '2022-07-14', '19 Cypress Ct', 'Chicago', 'IL',
'Midwest'),
(20, 'Tina', 'IT', 'Lead', 88000, 'F', '2016-10-31', '20 Juniper Way', 'Houston', 'TX',
'South'),
(21, 'Uma', 'HR', 'Manager', 60000, 'F', '2020-04-03', '21 Aspen Pl', 'Dallas', 'TX',
'South'),
(22, 'Victor', 'Finance', 'Analyst', 69000, 'M', '2018-07-19', '22 Cottonwood St', 'Dallas', 'TX',
'South'),
```

(23, 'Wendy', 'Marketing', 'Manager', 64000, 'F', '2021-11-27', '23 Palmetto Ave', 'Phoenix', 'AZ', 'West'),

(24, 'Xander', 'IT', 'Architect', 95000, 'M', '2015-02-14', '24 Mesquite Rd', 'Phoenix', 'AZ', 'West'),

(25, 'Yara', 'HR', 'Executive', 54000, 'F', '2022-09-08', '25 Saguaro Ln', 'Phoenix', 'AZ', 'West'),

(26, 'Zoe', 'Finance', 'Manager', 73000, 'F', '2019-05-21', '26 Palo Verde Dr', 'Dallas', 'TX', 'South'),

(27, 'Aaron', 'Marketing', 'Analyst', 67000, 'M', '2020-10-15', '27 Ocotillo Blvd', 'Phoenix', 'AZ', 'West'),

(28, 'Bella', 'IT', 'Senior Dev', 80000, 'F', '2017-08-06', '28 Ironwood St', 'Chicago', 'IL', 'Midwest'),

(29, 'Carlos', 'HR', 'Executive', 56000, 'M', '2021-06-24', '29 Pecan Ave', 'Houston', 'TX', 'South'),

(30, 'Diana', 'Finance', 'Lead', 77000, 'F', '2018-02-09', '30 Peachtree Ct', 'New York', 'NY', 'East'),

(31, 'Ethan', 'IT', 'Developer', 74000, 'M', '2019-04-18', '31 Magnolia St', 'Dallas', 'TX', 'South'),

(32, 'Fiona', 'Marketing', 'Executive', 61000, 'F', '2022-01-05', '32 Bluebonnet Ave', 'Dallas', 'TX', 'South'),

(33, 'George', 'Finance', 'Analyst', 66000, 'M', '2020-08-11', '33 Sunflower Rd', 'Phoenix', 'AZ', 'West'),

(34, 'Helen', 'HR', 'Manager', 59000, 'F', '2017-03-22', '34 Cactus Ln', 'Phoenix', 'AZ', 'West'),

(35, 'Ivan', 'IT', 'Lead', 92000, 'M', '2015-09-30', '35 Desert Dr', 'Phoenix', 'AZ', 'West'),

(36, 'Julia', 'Marketing', 'Manager', 65000, 'F', '2019-12-03', '36 Mesa Blvd', 'Dallas', 'TX', 'South'),

(37, 'Kevin', 'Finance', 'Manager', 70000, 'M', '2021-07-20', '37 Prairie St', 'Chicago', 'IL', 'Midwest'),

(38, 'Laura', 'HR', 'Executive', 55000, 'F', '2022-10-14', '38 Lakeview Ave', 'Chicago', 'IL', 'Midwest'),

(39, 'Mark', 'IT', 'Developer', 76000, 'M', '2018-05-07', '39 Riverview Rd', 'New York', 'NY', 'East'),

(40, 'Nina', 'Marketing', 'Analyst', 63000, 'F', '2020-03-25', '40 Bayview Ln', 'New York', 'NY', 'East'),

(41, 'Oscar', 'Finance', 'Analyst', 67000, 'M', '2019-08-19', '41 Hillside Dr', 'Dallas', 'TX', 'South'),

(42, 'Priya', 'HR', 'Senior Dev', 79000, 'F', '2016-11-28', '42 Creekside Blvd', 'Houston', 'TX', 'South'),

(43, 'Raj', 'IT', 'Manager', 85000, 'M', '2018-06-14', '43 Brookside St', 'Phoenix', 'AZ', 'West'),

(44, 'Sarah', 'Marketing', 'Lead', 71000, 'F', '2017-02-01', '44 Springdale Ave', 'New York', 'NY', 'East'),

(45, 'Tom', 'Finance', 'Senior Dev', 81000, 'M', '2020-09-09', '45 Fairview Rd', 'Chicago', 'IL', 'Midwest'),

(46, 'Uma', 'IT', 'Architect', 97000, 'F', '2014-12-22', '46 Northgate Ln', 'New York', 'NY', 'East'),

(47, 'Vera', 'Marketing', 'Manager', 68000, 'F', '2021-04-30', '47 Eastwood Dr', 'Dallas', 'TX', 'South'),

(48, 'Will', 'HR', 'Analyst', 51000, 'M', '2022-11-15', '48 Westfield Blvd', 'Phoenix', 'AZ', 'West'),

(49, 'Xena', 'Finance', 'Lead', 78000, 'F', '2019-10-06', '49 Southpark St', 'Chicago', 'IL', 'Midwest'),

(50, 'Yusuf', 'IT', 'Senior Dev', 87000, 'M', '2016-07-17', '50 Central Ave', 'Houston', 'TX', 'South');

select * from employees;

-- Queries 1-50

-- Q1. Find the total number of employees in each department

```sql
select dept,count(*) as totalemployees from employees group by dept;

-- Q2. Find the total salary paid per department.
select dept, sum(salary) as totalsalary from employees group by dept;

-- Q3. Find the average salary by gender.
select gender, avg(salary) as avgsalary from employees group by gender;

-- Q4. Find the maximum salary in each department.
select dept, max(salary) as maxsalary from employees group by dept;

-- Q5. Find the minimum salary in each role.
select role, min(salary) as minsalary from employees group by role;

-- Q6. count the number of employees in each city.
SELECT city, COUNT(*) AS empcount FROM employees GROUP BY city;

 -- Q7. find the total salary paid per region.
select region, sum(salary) AS Totalsalary from employees
group by region;

-- Q8. List department sorted by their average salary(highest first).
SELECT dept, AVG(salary) AS avgsalary from employees
group by dept
order by avgsalary desc;

-- Q9. List city by total salary ascending.
select city,SUM(salary) AS Totalsalary
from employees
group by city
order by Totalsalary ASC;
```

-- Q10. List roles sorted by employees count descending, then role name ascending.

```
select role, count(*) AS Empcount from employees
group by role
order by Empcount desc, role ASC;
```

-- Q11. List states with their employee count, sorted by count descending then state name.

```
select state, count(*) AS empcount from employees
group by state
order by empcount DESC, state ASC;
```

-- Q12. Show the total salary per region.

```
select region, sum(salary) AS Totalsalary from employees
group by region
order by Totalsalary desc;
```

-- Q13 list departments sorted by minimum salary descending.

```
select dept, min(salary) AS minsalary from employees
group by dept
order by minsalary desc;
```

-- Q14 show only departments with more than 10 employees.

```
select dept, count(*) AS empcount from employees
group by dept
having count(*) > 10;
```

-- Q15 show departments where the average saslary exeeds 65000.

```
select dept, AVG(salary) AS avgsalary from employees
group by dept
having avgsalary > 65000;
```

```sql
-- Q16 show cities where total salary exceeds 600000.
select city, SUM(salary) AS Totalsalary from employees
group by city
having SUM(salary) > 600000;
```

```sql
-- Q17 show roles with more than 7 employees
select role,count(*) AS empcount from employees
group by role
having count(*) > 7;
```

```sql
-- Q18 show regions where the average salary is below 70000;
select region, AVg(salary) AS avgsalary from employees
group by region
having avg(salary) < 70000;
```

```sql
-- Q19 show departments where the max salary is at least 90000.
select dept, max(salary) AS maxsalary from employees
group by dept
having max(salary) >= 90000;
```

```sql
-- Q20 show states where total headcount is more than 12.
select state, count(*) AS empcount from employees
group by state
having count(*)  > 12;
```

```sql
-- Q21 show cities where the minimum salary is above 60000.
SELECT City, MIN(Salary) AS MinSalary
FROM Employees
GROUP BY City
HAVING MIN(Salary) > 60000;
```

```
-- Q 22 count employees grouped by department AND city
select dept, city, count(*) AS empcount from employees
group by dept, city;

-- Q23 find average salary grouped by department AND gender.
select dept, gender, AVG(salary) AS avgsalary from employees
group by dept, gender;

-- Q24 count employees grouped by region AND role.
select region, role, count(*) AS empcount from employees
group by region, role;

-- Q25 find max salary grouped by state AND department.
select state, dept, max(salary) AS maxsalary from employees
group by state, dept;

-- Q26 find total salary grouped by city AND gender.
select city, gender, sum(salary) AS totalsalary from employees
group by city, gender;

-- Q27 count employees per department, city, and show groups with more that 2 employees.
select dept, city, count(*) AS empcount from employees
group by dept, city
having count(*) > 2;

-- Q28 find average salary grouped by region AND gender ordered by region and avg salary descending.
select region, gender, AVG(salary) AS avgsalary from employees
group by region, gender
order by region ASC, Avgsalary desc;
```

-- Q29 count employees grouped by state AND role, show only combinations with more than 1 employee.

select state, role, count(*) as empcount from employees

group by state, role

having count(*) > 1;

-- Q30 count how many employees joined in each year.

select extract(year from joindate) AS joinyear,count(*) AS joiners

from employees

group by extract(year from joindate)

order by joinyear;

-- Q31 find average salary of employees grouped by the year they joined.

select extract(year from joindate) as joinyear,

avg(salary) as avgsalary from employees

group by extract(year from joindate)

order by joinyear;

-- Q32 count how many employees joined in each month.

select extract(month from joindate) as joinmonth from employees

group by extract(month from joindate)

order by joinmonth;

-- Q33 find total salary grouped by quarter of the joindate.

select extract(quarter from joindate) as joinQTR,

sum(salary) AS Totalsalary from employees

group by Extract(quarter from joindate)

order by joinqtr;

-- Q34 count employees who joined in Q1(jan-mar) versus Q#(jul-sep).

select extract(quarter from joindate) as qtr, count(*) as joiners from employees

```sql
where extract(quarter from joindate) IN(1,3)
group by extract(quarter from joindate)
order by qtr;

-- Q35 show the year and department with head count, ordered by year and dept.
select extract(year from joindate) as joinyear,
dept, count(*) as empcount from employees
group by extract(year from joindate), dept
order by joinyear, dept;

-- Q36 show years where more than 6 employees joined.
select extract(year from joindate) AS joinyear,
count(*) AS joiners from employees
group by extract(year from joindate)
having count(*) > 6
order by joinyear;

-- Q37 find the average salary for employees joining in each quarter, show only quarters with avg salary above 70000.
select extract(quarter from joindate) AS qtr,
avg(salary) AS avgsalary from employees
group by extract(quarter from joindate)
having  avg(salary) > 70000
order by qtr;

-- Q38 find the number of employees and average salary per state.
select state, count(*) AS empcount, avg(salary) AS avgsalary
from employees
group by state
order by state;
```

```
-- Q39 find the department with highest headcount in each state.
select state, dept, count(*) AS empcount
from employees
group by state, dept
order by state, empcount desc;

-- Q40 show cities with more than 2 departments represented.
select city, count(distinct dept) AS deptcount from employees
group by city
having count(distinct dept) > 2;

-- Q41 find the total salary bill per region AND state.
select region, state, sum(salary) AS totalsalary
from employees
group by region, state
order by region, state;

-- Q42 show your region with more than 10 employees AND average salary above 67000,
select region, count(*) AS empcount, avg(salary) AS avgsalary
from employees
group by region
having count(*) > 10
AND avg(salary) > 67000;

-- Q43 count distinct roles available in each city.
select city, count(distinct role) AS uniquerole
from employees
group by city
order by city;

-- Q44 count employees per department, but only consider female employees.
```

```sql
select dept, count(*) AS femalecount
from employees
where gender = 'f'
group by dept;

-- Q45 find average salary per department for employees who joined after 2019-01-01.
select dept, avg(salary) AS avgsalary
from employees
where joindate > '2019-01-01'
group by dept;

-- Q46 find total salary per city from employees in the IT department only.
select city, sum(salary) AS Totalsalary from employees
where dept = 'IT'
group by city;

-- Q47 count employees per role for those with salary above 70000.
select role, count(*) AS empcount
from employees
where salary > 70000
group by role;

-- Q48 show the max salary per department, only for employees in the south or west region.
select dept, max(salary) AS maxsalary
from employees
where region IN ('south', 'west')
group by dept;

-- Q49 count employees per year of joining for texas(TX) state only.
select extract(year from joindate) AS joinyear,
count(*) AS empcount from employees
```

```sql
where state = 'TX'
group by extract(year from joindate)
order by joinyear;

-- Q50 find department where more than 3 female employees have salaries above 60000.
select dept, count(*) AS eligiblefemales
from employees
where Gender = 'F'
AND salary > 60000
group by dept
having count(*) > 3;
```